

Ishan Paharia

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EDUCATION

The LNM Institute of Information Technology

B.Tech in Computer Science, CGPA: 8.22/10

Jaipur, India

2023 – 2027

PROJECTS

Pixnette — Real-Time Collaborative Pixel Canvas — Live Demo

React, Tailwind CSS, Node.js, Express, PostgreSQL (Neon), Socket.IO, Redis, Docker, Nginx

- Engineered horizontal scaling across 2 Node.js instances behind an Nginx round-robin reverse proxy, sustaining ~300 concurrent WebSocket connections on a self-hosted VPS.
- Built a Redis write-back pipeline that batches up to 4,000 pixel events per PostgreSQL transaction, reducing write amplification by ~90% while preserving full timelapse history.
- Synchronized multi-instance Socket.io state via `@socket.io/redis-adapter` Pub/Sub and implemented a 4 KB binary Redis canvas using `SETRANGE/GET` for O(1) per-pixel reads and writes.
- Hardened deployment with Redis TTL cooldowns, UUID/IP-UA fingerprinting, Nginx rate limits, and GitHub Actions CI/CD for automated Docker deployments.

CoursessionAI — AI Learning Platform — Live Demo

React, Tailwind CSS, Node.js, Express, PostgreSQL (Neon)

- Built a full-stack platform that converts YouTube playlists into structured AI-generated courses with module hierarchies, summaries, quizzes, and a context-aware AI tutor.
- Engineered an atomic ingestion pipeline executing 50+ relational writes within a single transaction, eliminating partial-state corruption across course, module, and video creation.
- Reduced authentication database lookups by up to 95% using a 5-minute in-memory cache, while enforcing Clerk JWT authentication and resource-level ownership checks across 40+ protected API endpoints.
- Optimized YouTube playlist ingestion by batching metadata requests in groups of 50 videos, reducing external API calls by up to 98%, and deployed on Vercel with rate-limited APIs.

DFilesS — Distributed File Store — Live Demo

React, Tailwind CSS, TypeScript, Node.js, Fastify, PostgreSQL, Docker

- Designed a distributed file storage system that splits files into checksummed 4 MB chunks replicated across 3 storage nodes with a 2-of-3 write quorum for fault-tolerant persistence.
- Implemented heartbeat-based failure detection every 5 seconds with automatic replica failover, maintaining read availability during single-node failures.
- Built a self-healing repair loop that restores replication every 7.5 seconds by verifying SHA-256 checksums and transferring under-replicated chunks between storage nodes.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript

Frameworks: React, Next.js, Node.js, Express, Fastify, Django, FastAPI, Tailwind CSS

Databases: MongoDB, PostgreSQL, MySQL, Redis

Tools & Technologies: Docker, WebSockets, Git, Linux/Bash

Others: Data Structures & Algorithms, OOP, DBMS

CO-CURRICULAR

Coordinator and Mentor

Aug 2024 – Apr 2026

Art and Craft Club (Vignette)

- Coordinated workshops and events for 100+ participants, handling end-to-end logistics and scheduling.
- Mentored junior members across semester-long creative projects, driving consistent participation.

ACHIEVEMENTS

Codeforces: Specialist (Max Rating: **1468**)

CodeChef: 3 Star Coder (Max Rating: **1746**)

LeetCode: Knight (Max Rating: **1919**)

Problem Solving: Solved **700+** DSA/CP problems across multiple platforms